

Bot 2102 Short Questions

1. Clarify the shape of eukaryotic cell.

- Cell shapes are almost as numerous as cell types: there is not typical shape.
- The cells of certain unicellular forms, such as *Amoeba*, *Diatoms*, *Acetabularia* and bacteria exhibit a number of shapes.
- But generally, the cells are rounded or spherical. Besides this the cells like oval, cuboidal, cylindrical, flat, discoid, polygonal have also been observed.
- The shape of cells depends mainly on functional adaptations and partly on the surface tension and viscosity of the protoplasm, the mechanical action exerted by the adjoining cells and the rigidity of the cell membrane.
- Some cells, such as *Amoeba* and leucocytes, can change their body shape very frequently.

2. Mention the function of cell wall.

- The function of cell wall in plant is believed mainly to provide mechanical strength.
- Particularly in higher plants having vascular system, cell walls provide main supporting framework.
- Since in animal some sort of exoskeleton or endoskeleton is present, perhaps there is hardly any necessity of definite cell walls.
- Therefore, besides others, main function of the cell wall is the same as that of a skeleton in animals.

3. Write short notes the nucleolus.

- The nucleus contains a large, spherical and acidophilic dense known as the nucleolus. The nucleoli were discovered by Fontana in 1781.
- It was well established in 19th century that the size of the nucleolus is related with the synthetic activity of the cell.
- Therefore, the cell with little or no synthetic activities, e.g., sperm cells, blastomeres, muscle cell, etc., are found to contain smaller or no nucleoli, while the oocytes, neurons and secretory cells which synthesize the proteins or other substances contain comparatively large sized nucleoli.
- The number of the nucleoli in the nucleus depends on the species and the number of the chromosomes. The number of the nucleoli in the cell may be one, two or four.
- The position of the nucleolus in the nucleus is eccentric.

Bot 2102 Long Question

1. Explain in detail the plasma membrane.

- The outer most boundary of the animal cell is called the 'plasma membrane'.
- It is invisible under a light microscope but under electron microscope, it appears to be composed of two dense layers.
- Both these layers are about $20\text{\AA}$ in thickness, and are separated from each other by a less dense area of about $35\text{\AA}$ in thickness.
- By this way the total thickness of plasma membrane is about $75\text{\AA}$.
- Before the advance of electron microscope people used to think of the plasma membrane has been stretched tightly over the cell.
- But now we know that a single plasma membrane can have as many as 3000 microvilli another peculiarity of the structure of the membrane is that, it remains connected with the endoplasmic reticulum. $\rightarrow$ 1.4.1 pg 9
- Plasma membrane is lipo-proteinaceous in composition.

Functions

- (i) Gives an identity to the cell.
- (ii) Protects the cell organelles.
- (iii) Helps in digestion through phagocytosis and pinocytosis.
- (iv) Being permeable to water, water can pass through it.
- (v) Help in the transport of ions.
- (vi) Helps in the formation of cytoplasmic organelles.